

Teleport Protocol Specification

Omnichain Collateralized Credit and Settlement

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System Definition

Teleport is an **omnichain collateralized credit and settlement protocol**. It is not a bridge, vault, stablecoin, or liquid staking token.

The system comprises four subsystems, each audited against different failure classes:

1. **Collateral custody:** Vault contracts holding base assets (ETH, BTC, etc.)
2. **Credit issuance:** Minting/burning credit tokens (LETH, LBTC) against collateral
3. **Transport/finality:** Cross-chain message passing with MPC attestation
4. **Surplus allocation:** Yield distribution to protocol equity holders (xLUX)

Asset Taxonomy

Class 1: Credit Assets

Examples: LETH, LBTC

A credit asset is an overcollateralized, in-kind redeemable liability of the protocol, minted against vault-backed base assets subject to a hard LTV cap.

Required properties:

- Minted only against recognized backing
- Redeemable 1:1 in kind
- Hard issuance cap at or below configured LTV
- No direct claim on strategy yield
- No governance, profit-share, or security rights

Constraint (enforced technically and legally):

$$\text{creditAsset} \neq \text{yieldShare} \neq \text{securityToken}$$

Enforcement:

- Separate token contracts
- Separate registries
- Separate accounting buckets
- Separate docs and disclosures

Canonical definition of LETH:

LETH is a protocol-issued, fully collateralized, in-kind redeemable credit token backed by ETH vault assets and minted subject to a hard maximum loan-to-value ratio of 90%, with strategy yield accruing to protocol surplus rather than to LETH holders.

Class 2: Yield/Equity Instruments

Examples: xLUX, protocol equity accounting unit

A yield/equity instrument represents a claim on protocol surplus, not on redemption principal backing.

Required properties:

- Absorbs upside from strategy earnings
- May absorb losses after buffer rules, depending on design
- Never used as collateral equivalence for redemptions unless explicitly approved

Class 3: Security Assets

Examples: tokenized equity, funds, RWAs

A security asset is issued and controlled under the ATS/compliance layer and is not fungible with protocol credit liabilities.

Required properties:

- Compliance-enforced transferability
- Lockup / whitelist logic
- Jurisdictional controls
- Separate wrapping policy when bridged into Lux

Bridge Message Schema

Message Types

Type	Purpose
DEPOSIT_V1	Lock base asset, request credit mint on destination
BURN_V1	Burn credit token, request base asset release
RELEASE_V1	Release base asset from vault
BACKING_V1	Attest current backing state

Common Envelope

```

MessageEnvelope {
  version:          uint8
  messageType:      bytes4
  srcChainId:       uint256
  dstChainId:       uint256
  routerAddress:    address
  assetId:          bytes32
  nonce:            uint256
  amount:           uint256
  recipient:        address
  expiry:           uint256
  originTxHash:     bytes32    // source event identifier

```

```

    payloadHash:    bytes32    // canonical hash of typed payload
    domainSeparator: bytes32    // chain + deployment + protocol version
}

```

The `domainSeparator` closes replay classes:

- Chain replay
- Deployment replay
- Router migration replay
- Same-signature / different-context reuse
- Duplicated source event processing

Verification Rules

Every receive path MUST check:

1. `block.timestamp <= expiry`
2. `dstChainId == currentChainId`
3. `routerAddress == localRouter`
4. Nonce unused for `(srcChainId, routerAddress, assetId)`
5. `amount > 0`
6. `recipient != address(0)`
7. `assetId` recognized and enabled
8. Message hash signed under current authorized signer set

Invariants

Definitions

Let:

- B = total backing value of redeemable vault assets
- L = total outstanding credit liabilities
- $E = B - L$ = protocol equity

Core Issuance Invariant

$$L \leq \text{issuanceLtvCap} \times B$$

For LETH at 90% max LTV:

$$L \leq 0.9 \times B$$

Redemption Invariant

$$\text{burn}(x \text{ LETH}) \Rightarrow \text{release}(x \text{ ETH})$$

Subject only to:

- Finality delay (per-chain confirmation threshold)
- Operational pause (emergency state only)
- Sanctions/compliance constraints where applicable

Parity Invariant

$$\text{redeemable principal per unit of LETH} = 1 \text{ ETH}$$

Equity Ratio

$$\frac{E}{B} \geq 10\%$$

Equivalently under a 90% LTV model:

$$\frac{L}{B} \leq 90\%$$

Solvency State Machine

States

State	Condition	Description
HEALTHY	$B \geq L/0.9$ and $B \geq L$	Normal operation
RESTRICTEDMINT	$B < L/0.9$ but $B \geq L$	Overcollateralization violated, principal intact
EMERGENCY	$B < L$	Undercollateralized on principal basis
RECOVERY	Governance-initiated	Recapitalization in progress

Auto-pause threshold:

$$B < 0.985 \times L \Rightarrow \text{EMERGENCY}$$

Allowed Transitions

HEALTHY

- > RESTRICTED_MINT (B drops below L/0.9)
- > EMERGENCY (B drops below L, or $B < 0.985 \times L$)

RESTRICTED_MINT

- > HEALTHY (B recovers above L/0.9)
- > EMERGENCY (B drops below L)

EMERGENCY

- > RECOVERY (governance vote or MPC action)

RECOVERY

- > HEALTHY (recapitalization complete, invariants restored)

Per-State Permissions

Action	Healthy	Restricted	Emergency	Recovery
Mint credit tokens	✓	×	×	×
Burn credit tokens	✓	✓	✓	✓
Release base assets	✓	✓	Queue*	Queue*
Strategy deposits	✓	×	×	×
Strategy withdrawals	Normal	Accelerated	Forced	Forced
ATS wrapping	✓	✓	×	×

*In Emergency/Recovery, releases enter a FIFO queue processed as backing becomes available.

State Transition Authority

- HEALTHY → RESTRICTEDMINT: Automatic (contract-enforced)
- RESTRICTEDMINT → EMERGENCY: Automatic (contract-enforced)
- EMERGENCY → RECOVERY: MPC quorum (T-of-N threshold)
- RECOVERY → HEALTHY: MPC quorum + invariant verification

Failure Behavior

MPC Halt

Mode B (user-executable fallback):

- After timeout T , users can prove burn on source chain and self-relay on destination
- Release executes from pre-authorized fallback path
- No liveness dependency on MPC signer availability

This provides institution-grade guarantees: user funds are never permanently locked by signer unavailability.

Stale Backing Report

$$\text{maxBackingReportAge} = X \text{ seconds}$$

Policy:

- If stale: disable mint
- Allow burn and release against last confirmed backing unless in EMERGENCY state
- Emit `BackingStale` status event

Asymmetry: stop issuance first, preserve exits as long as possible.

Strategy Loss

Loss waterfall:

1. Realized/unrealized loss hits surplus buffer (E)
2. If buffer exhausted: system enters RESTRICTEDMINT

3. If principal backing threatened: EMERGENCY
4. New deployment to strategies halted until recapitalization

Chain Reorg / Finality

Per chain, define:

`minFinalityConfirmations[chainId]`

Mint/release occurs only after finality threshold for source events. No global constant across chains.

Oracle / Valuation Failure

Policy:

- Freeze mint for affected asset
- Allow only like-for-like redemption where backing is directly observable
- Disable cross-asset accounting based on stale prices

ATS Boundary

A security token may be transported into Lux only through a wrapper whose compliance posture is explicitly declared and enforced at the wrapper level.

Strict Wrapper Model (Recommended)

`ATS_SECURITY → Restricted_Wrapped_Security`

Properties preserved:

- Whitelist enforced
- Transfer restrictions preserved
- Lockups preserved
- DeFi composability limited by design

The dual wrapper model (restricted → composable) is deferred until policy and counsel are settled. Start strict.

Trust Assumptions

Component	Trust Assumption
Vault contracts	Solidity correctness (auditable, immutable after deployment)
MPC signers	Honest T-of-N threshold (dishonest minority cannot forge)
Strategy yield	External protocol risk (Aave, Lido, etc.) bounded by allocation limits
Oracle/pricing	Freshness guarantee within <code>maxBackingReportAge</code>
Chain finality	Source chain provides irreversibility after <code>minFinalityConfirmations</code>
ATS compliance	Wrapper contracts enforce jurisdictional rules
Consensus	Lux triple-proof quantum finality (BLS + ZK/ML-DSA + Ringtail)

Recommended Sequencing

1. Freeze this protocol spec
2. Freeze the message schema
3. Freeze the solvency state machine
4. Map every state transition and invariant to contract checks
5. Deployment review and audit prep

The reason to spec first: line-by-line router review is much more valuable once the semantics are fixed.

Open Questions (Pre-Audit)

Two spots require exact wording before audit:

1. **Stale backing + redemptions:** Whether redemptions remain permissionless during stale-backing conditions.
Recommendation: stop mint, preserve exits.
2. **Emergency undercollateralization:** Whether releases pause entirely or shift to queue/pro-rata/governance-controlled mode.
Recommendation: stop mint, preserve claim ordering through an explicit emergency redemption policy (FIFO queue).

These are core user-rights definitions, not implementation details.